

DEEP MEHTA

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Portfolio | GitHub | LinkedIn | LeetCode | CodeChef

Education

Indian Institute of Information Technology, Vadodara (Gandhinagar Campus)

2024 – 2028

Bachelor of Technology in Computer Science Engineering (CGPA: 8.85/10)

Gujarat, India

Technical Skills

Languages: Java, C++, Python, JavaScript, SQL

Backend: Node.js, Express.js, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, pgvector

Frontend: React, HTML, CSS, Tailwind

AI/ML: RAG Systems, LLM Applications, Prompt Engineering, Embeddings, Vector Search

Tools: Git, GitHub, Linux, Postman, VS Code

Projects

Resyntha – AI-Powered Research Intelligence Platform

Ongoing

React, FastAPI, PostgreSQL, pgvector, LangGraph, Groq

- Building a multi-agent AI platform that retrieves, analyzes, and synthesizes scientific literature into structured research intelligence.
- Designing specialized agent workflows for paper retrieval, metadata extraction, comparative analysis, trend identification, and research opportunity discovery.
- Architecting an asynchronous, production-grade backend with FastAPI, PostgreSQL, pgvector, Redis, and LangGraph using scalable system design principles.

Atlas – Production Multi-Tenant Document RAG Platform

React, Node.js, PostgreSQL, pgvector, Better Auth, Groq, HuggingFace

- Built a production-grade Retrieval-Augmented Generation (RAG) platform supporting secure multi-user document ingestion, semantic retrieval, and citation-grounded AI conversations.
- Designed an end-to-end retrieval pipeline with PDF extraction, recursive chunking, BAAI/bge-small-en-v1.5 embeddings, pgvector HNSW indexing, semantic search, and LLM response generation.
- Architected a modular backend using dependency injection, repository-service pattern, streaming (SSE), and Better Auth with PostgreSQL-backed session management; deployed on Vercel, Render, and Neon.

Intelligent Engineering Portfolio AI Assistant

React, Node.js, Express.js, Groq API

- Built a portfolio website with an AI assistant capable of answering questions about projects, skills, and technical experience.
- Implemented retrieval-based context injection, backend APIs, and production deployment on Vercel.

Smart University Academic Placement System

Node.js, Express.js, PostgreSQL

- Designed a BCNF-normalized database supporting academic, grading, and placement workflows.
- Built role-based backend modules for Students, Faculty, and Administrators using PostgreSQL transactions and validations.

Achievements

- Solved **800+** DSA problems on **LeetCode** and achieved a **1600+** rating on **CodeChef** through competitive programming.
- DAIICT HackOut Finalist – 2025
- Joint Secretary, Encore – Music Club, IIIT Vadodara
- Languages: English, Hindi, Gujarati